Dr. Vishwanath Karad
MIT WORLD PEACE UNIVERSITY
PUNE, INDIA
TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

Dr. Vishwanath Karad MIT WORLD PEACE UNIVERSITY, PUNE

A MINI PROJECT REPORT ON

EXPLAINABLE AI-BASED DRUG DISCOVERY ASSISTANT FOR EARLY MOLECULAR SCREENING

Submitted in partial fulfillment of the requirements for the course
AI SYSTEMS AND APPLICATIONS

Submitted by:

Kuntal Saha (PRN / Roll No: _____)
[Team Member 2 Name] (PRN: _____)
[Team Member 3 Name] (PRN: _____)

Under the Guidance of:

Dr. Devendra Joshi
Department of Artificial Intelligence & Data Science,
School of Computer Science & Engineering

Department of Artificial Intelligence and Data Science
Dr. Vishwanath Karad MIT World Peace University
Paud Road, Kothrud, Pune, Maharashtra – 411 038, India

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ABSTRACT & INTRODUCTION

Traditional pharmaceutical drug discovery is an extraordinarily costly, labor-intensive, and protracted endeavor, typically spanning 10 to 15 years and requiring capital investments exceeding $2 billion per approved therapeutic entity. A predominant cause of this severe developmental attrition is the late-stage failure of candidate chemical compounds during preclinical animal trials and human clinical phases, largely attributable to inadequate target binding affinity, off-target cellular toxicity, or poor oral pharmacokinetics. In classical High-Throughput Screening (HTS), pharmaceutical laboratories empirically assay millions of physical chemical compounds against disease proteins, consuming vast quantities of expensive biological reagents and technician hours.

This mini-project for **AI Systems and Applications**, developed under the mentorship of **Dr. Devendra Joshi** at **MIT World Peace University (MIT-WPU), Pune**, introduces **EGFR Molecular Intelligence**—an end-to-end, interpretable computational framework for early-stage Virtual High-Throughput Screening (vHTS). The application specifically targets **Human Epidermal Growth Factor Receptor (EGFR / ErbB-1, ChEMBL ID: CHEMBL203)**, a transmembrane receptor tyrosine kinase whose hyper-activation drives oncogenic proliferation in Non-Small Cell Lung Cancer (NSCLC) and Glioblastoma.

By training on a curated benchmark of 881 small-molecule bioactivity assays extracted from the EMBL-EBI ChEMBL database, the platform translates raw chemical representations (SMILES notation) into composite 2,056-dimensional numerical feature spaces combining **2048-bit Extended Connectivity Circular Fingerprints (ECFP4)** with 8 physicochemical property descriptors. Supervised ensemble models—**Random Forest (RF)** and **Gradient Boosted Decision Trees (GBDT)**—achieved exceptional predictive performance on a holdout test set with test accuracies of **90.96%** and **91.53%**, and Area Under the ROC Curve (**ROC-AUC**) metrics of **0.978** and **0.968**, respectively.

To resolve the opaque 'black-box' dilemma that historically impedes clinician and chemist trust in biomedical AI, the platform incorporates **Explainable AI (XAI)** based on **TreeSHAP (SHapley Additive exPlanations)** cooperative game theory. The system computes exact polynomial-time Shapley attributions and projects them directly onto 2D molecular graphs, highlighting favorable binding anchors in fluorescent chartreuse and unfavorable steric penalties in crimson. The platform additionally assesses oral drug-likeness via **Lipinski’s Rule of Five** and **Veber’s rules**, computes **Tanimoto structural similarity** against FDA-approved cancer drugs (Gefitinib, Erlotinib, Osimertinib), embeds an offline **Ollama Large Language Model (LLM)** chemoinformatics tutor, and compiles automated vector PDF screening dossiers via ReportLab.

KEYWORDS: Explainable AI (XAI), Artificial Intelligence Systems, Virtual Screening, Epidermal Growth Factor Receptor (EGFR), Human ChEMBL203, SHAP TreeExplainer, RDKit, Random Forest, Gradient Boosting, Morgan Fingerprints (ECFP4), Lipinski Rule of Five, MIT-WPU Pune.

SECTION 1: MOTIVATION & PROBLEM STATEMENT

1.1 The Pharmaceutical Discovery Bottleneck

Pharmaceutical development faces steep attrition: only 1 in 10,000 synthesized compounds ultimately gains regulatory approval. Traditional High-Throughput Screening (HTS) requires hundreds of thousands of robotic wet-lab enzymatic assays, incurring substantial costs. Deploying **Narrow Artificial Intelligence as an intelligent pre-filter** enables rapid *in silico* digital evaluation, discarding non-viable chemical structures before physical synthesis occurs.

1.2 Target Biology: Oncogenic Human EGFR (CHEMBL203)

The Epidermal Growth Factor Receptor (EGFR / ErbB-1) is a transmembrane tyrosine kinase receptor. Somatic activating mutations—specifically exon 19 deletions and the L858R point mutation—drive uncontrolled proliferation in Non-Small Cell Lung Cancer (NSCLC). While 1st-generation ATP-competitive inhibitors (Gefitinib) provide clinical benefits, acquired secondary mutations (T790M) necessitate continuous discovery of diverse chemical scaffolds capable of kinase inhibition.

1.3 Key Project Objectives

• Build an end-to-end Python pipeline parsing SMILES strings into energy-minimized 3D conformers.
• Curate 881 bioactivity assays from ChEMBL (CHEMBL203) with a verified IC■■ ≤ 1,000 nM active threshold.
• Formulate 2,056-dimensional feature spaces (2,048 ECFP4 bits + 8 RDKit descriptors).
• Train Random Forest and GBDT ensembles achieving ROC-AUC > 0.96.
• Provide atom-level TreeSHAP game-theoretic explainability on 2D diagrams.
• Profile oral bioavailability (Lipinski/Veber) and clinical Tanimoto similarity.
• Deliver an interactive Streamlit UI, an offline Ollama LLM tutor, and automated PDF reporting.

SECTION 2: END-TO-END SYSTEM WORKFLOW

2.1 Pipeline Flowchart & Architectural Workflow

The system operates as an integrated multi-stage inference and explainability engine:

STAGE 1: Chemical Ingestion & Geometry Optimization
User SMILES ■■■ RDKit Validation ■■■ 2D Planar SVG ■■■ ETKDGv3 MMFF94 3D Conformer

STAGE 2: Multi-Scale Molecular Featurization
2048-bit Morgan Fingerprints (ECFP4, r=2) + 8 Scaled Physicochemical Descriptors ■■■ Feature Vector x in R^2056

STAGE 3: Supervised Ensemble Inference & Probability Calibration
Random Forest (100 Trees) & GBDT (100 Estimators) ■■■ P(Active) ■■■ Decision Threshold (0.50)

STAGE 4: Game-Theoretic Explainability (TreeSHAP)
Exact Polynomial TreeSHAP ■■■ RDKit bitInfo Coordinate Mapping ■■■ 2D Atom Highlight Map

STAGE 5: Translational Intelligence & Automated Reporting
Lipinski/Veber Radar Chart ■■■ Tanimoto Similarity vs FDA Drugs ■■■ Ollama LLM Tutor ■■■ PDF Export

SECTION 3: SYSTEM METHODOLOGY

3.1 ChEMBL Data Curation & Logarithmic Transformation

Bioactivity records for target CHEMBL203 were filtered for IC■■ measurements in nM and transformed to logarithmic potency: **pIC■■ = 9 - log■■(IC■■)**. Compounds with IC■■ ≤ 1,000 nM (pIC■■ ≥ 6.0) were labeled Active (y=1), and IC■■ ≥ 10,000 nM were labeled Inactive (y=0). Ambiguous intermediates were excluded, yielding 881 curated compounds.

3.2 Featurization & Physicochemical Descriptors

Each molecule is transformed into 2048-bit circular Morgan fingerprints (ECFP4, radius 2) and 8 standardized bulk descriptors:

Descriptor	Physical Property	Clinical Role in Drug Discovery
MolWt	Molecular Weight (Da)	Cavity fit, passive transcellular diffusion
MolLogP	Octanol-water partition	Lipophilicity, gut absorption, membrane permeability
TPSA	Polar Surface Area (Å²)	Epithelial permeability, blood-brain barrier transport
HBD / HBA	H-Bond Donors & Acceptors	Directional binding capability with ATP-pocket residues
NumRotBonds	Rotatable Bond Count	Conformational flexibility; high counts penalize entropy
QED	Drug-likeness desirability	Desirability metric combining 8 properties (0.0 to 1.0)

3.3 TreeSHAP Game-Theoretic Explainability

Model predictions are decomposed into additive Shapley contributions: **f(x) =** ϕ■ **+** Σ ϕ■**(x)**. Active fingerprint bits are mapped to central atoms and neighborhood radii via RDKit bitInfo, yielding color-coded 2D atomic highlights (Chartreuse = Favorable binding anchor, Crimson = Steric penalty).

SECTION 4: EXPERIMENTAL RESULTS & DISCUSSION

4.1 Holdout Test Set Performance Metrics

Evaluated on 177 holdout compounds (20% stratified test split from 881 curated ChEMBL molecules):

Model Classifier	Test Accuracy	Precision	Recall	F1-Score	Test ROC-AUC
Random Forest (RF)	**90.96%**	90.62%	92.55%	0.9158	**0.9784**
Gradient Boosting (GBDT)	**91.53%**	91.67%	92.55%	0.9206	**0.9678**

4.2 Explainable AI Case Studies (Gefitinib vs. Aspirin Control)

• **Gefitinib (Approved Active):** Predicted Active with **99.8% probability**. Top positive SHAP drivers identified the 4-anilinoquinazoline core and the 3-chloro-4-fluoroaniline moiety, matching crystallographic hinge-binding anchors with Met793.
• **Aspirin (Negative Control):** Predicted Inactive with **7.7% probability**. SHAP identified low molecular weight (180 Da) and the negatively charged carboxylate group as severe penalties (-SHAP), causing electrostatic repulsion with the hydrophobic ATP pocket.

4.3 Oral Bioavailability & Clinical Similarity Benchmarks

Gefitinib and Erlotinib pass all Lipinski and Veber criteria with 0 violations. Tanimoto similarity against clinical drugs correctly identified Gefitinib at 100%, Erlotinib at 100%, and Canertinib at 78.4% homology.

SECTION 5: CONCLUSION & FUTURE SCOPE

5.1 Conclusion

This mini-project for **AI Systems and Applications** at **MIT World Peace University (MIT-WPU), Pune**, conducted under the guidance of **Dr. Devendra Joshi**, successfully established an explainable virtual screening platform for Human EGFR inhibitors. Combining 2048-bit Morgan circular fingerprints with tree ensembles yielded **90.96% accuracy and 0.978 ROC-AUC**, while TreeSHAP provided actionable atom-level structural rationales.

5.2 Key Technical Advantages

- **Sub-Second Throughput:** Screens molecules *in silico* in < 50 ms, saving months of wet-lab screening.
- **Actionable Interpretability:** Decomposes black-box predictions into atom-level highlights for lead optimization.
- **Dual Affinity & Bioavailability Filtering:** Concurrently evaluates binding and Lipinski/Veber oral absorption.
- **Offline Privacy:** Local Ollama LLM and Python inference ensure zero external cloud data exposure.

5.3 Future Scope

- Extend featurization to 3D Equivariant Graph Neural Networks (EGNN).
- Expand models across the multi-target human kinome (HER2, HER4, VEGFR2).
- Couple with reinforcement learning for *de novo* drug design.
- Incorporate quantitative hERG cardiac and CYP450 toxicity predictors.

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